Learning Objective 6.4: I can implement an mPSK modulator and demodulator (BPSK, QPSK, 8PSK), recover carrier and symbol timing, and verify constellation behavior against theory.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Bits per symbol.** A telemetry radio sends 25,000 symbols per second with RRC pulses of roll-off $\alpha = 0.25$. Lesson 3 gives the E_b/N_0 each modulation needs for a bit error rate of 10^{-5}: 9.6 dB for BPSK and QPSK, and 13.0 dB for 8PSK.

   (a) Find the occupied bandwidth, and the bit rate with BPSK, QPSK, and 8PSK.

   B: 31.25 kHz

   $$B = (1+\alpha)R_s = 1.25 \times 25\ \text{kHz} = 31.25\ \text{kHz}$$

   The bandwidth does not depend on M. The bit rate is kR_s with $k = \log_2 M$: 25, 50, and 75 kb/s.

   bits: 25, 50, 75 kb/s

   (b) The link budget delivers $E_s/N_0 = 15$ dB at the receiver. Find E_b/N_0 for each modulation, and say which ones reach a bit error rate of 10^{-5}.

   BPSK and QPSK

   $E_b/N_0 = E_s/N_0 - 10\log_{10} k$:

   BPSK: $15 - 0 = 15.0$ dB
   QPSK: $15 - 3.0 = 12.0$ dB
   8PSK: $15 - 4.8 = 10.2$ dB

   BPSK and QPSK clear 9.6 dB. 8PSK needs 13.0 dB and falls 2.8 dB short.

   (c) In words, explain why this radio would use QPSK rather than BPSK, and what it would give up by moving to 8PSK.

   QPSK needs the same E_b/N_0 as BPSK for the same error rate and sends twice the bits in the same 31.25 kHz, so BPSK has no advantage here. 8PSK would send 50% more bits again in the same band, but it needs 3.4 dB more E_b/N_0, and this link does not have it: the error rate would rise well above 10^{-5}.

2. **The Gray map.** The L15 transmitter numbers the QPSK points by *phase index* $i = 0, 1, 2, 3$, counterclockwise from the point at $45°$. The Gray label of index i is $i \oplus (i \gg 1)$, the index XORed with itself shifted right one bit.

   (a) Write the 2-bit Gray label of each index. Then find the indices the transmitter sends for the bits `01 11 10 00`.

   $$0 \oplus 0 = 00, \quad 1 \oplus 0 = 01$$
   $$2 \oplus 1 = 11, \quad 3 \oplus 1 = 10$$

   The labels are 00, 01, 11, 10. The transmitter's Map sends the index whose label matches each pair: `01` is 1, `11` is 2, `10` is 3, and `00` is 0, so it sends 1, 2, 3, 0.

   labels: 00, 01, 11, 10

   indices: 1, 2, 3, 0

   (b) Noise moves a symbol sent as index 1 into the region of index 2. How many bits are wrong with the Gray labels? How many would be wrong if index i simply carried the binary number i?

   Gray: `01` becomes `11`, one bit. Natural binary: `01` becomes `10`, two bits.

   Gray: 1

   binary: 2

   (c) In words, explain why the BER of a Gray-coded 8PSK link is close to its symbol error rate divided by 3.

   Almost every symbol error is a slip to one of the two neighbors, because the noise rarely reaches farther. With Gray labels a neighbor differs in one of the three bits, so each symbol error corrupts about one bit out of three, and the bit error rate is about a third of the symbol error rate.

3. **Differential coding by hand.** The L15 Differential Encoder sends $e_n = (e_{n-1} + d_n) \bmod M$, where d_n is the phase index from the Gray map and e_{n-1} starts at 0. The decoder outputs $(r_n - r_{n-1}) \bmod M$, where r_n is the received index and r_{n-1} also starts at 0. The link uses QPSK, and the data indices are $d = 2, 1, 1, 3, 0$.

   (a) Find the five indices the encoder sends.

   $$e_0 = 0 + 2 = 2, \quad e_1 = 2 + 1 = 3$$
   $$e_2 = (3 + 1) \bmod 4 = 0$$
   $$e_3 = 0 + 3 = 3, \quad e_4 = 3 + 0 = 3$$

   2, 3, 0, 3, 3

   (b) The receiver's Costas Loop locks 180° off, so every received index is 2 more than the one sent, mod 4. Decode the five received indices and compare them with d.

   Received: $r = 0, 1, 2, 1, 1$. Decoded, with $r_{-1} = 0$:

   $$0 - 0 = 0,\ 1 - 0 = 1,\ 2 - 1 = 1,\ 1 - 2 \equiv 3,\ 1 - 1 = 0$$

   0, 1, 1, 3, 0

   The output is 0, 1, 1, 3, 0. Every symbol after the first is right; the first is wrong only because the decoder's starting reference was not rotated.

   (c) Now the loop locks correctly, but noise turns the third received index from 0 into 1. Decode the five indices and count the wrong symbols.

   Received: $r = 2, 3, 1, 3, 3$.

   $$2,\ 3 - 2 = 1,\ 1 - 3 \equiv 2,\ 3 - 1 = 2,\ 3 - 3 = 0$$

   2 wrong

   The output is 2, 1, 2, 2, 0: the third and fourth symbols are wrong, two errors from one wrong decision.

   (d) In words, use parts (b) and (c) to explain what differential coding does for a link and what it does to the bit error rate.

   It makes the link work at any of the M lock positions, because a rotation adds the same amount to every index and the differences do not change. In exchange, every wrong decision spoils two differences, the one into it and the one out of it, so the bit error rate roughly doubles, 0.3 to 0.4 dB in Lesson 15's measurements.

4. **The frequency offset.** A frequency offset Δf turns the constellation by $360^\circ \times \Delta f / R_s$ every symbol. Lesson 15 measured the pull-in range of the QPSK Costas Loop at about 375 Hz for $R_s = 4{,}000$ symbols per second and an E_b/N_0 of 12 dB. The loop works once per symbol, so with the same loop settings its pull-in range scales with the symbol rate.

   (a) A 4 kbaud QPSK link has a 75 Hz offset. Find the rotation per symbol and the number of symbols in one full turn of the constellation.

   $$360^\circ \times \frac{75}{4000} = 6.75^\circ$$

   $$\frac{360^\circ}{6.75^\circ} = 53.3 \text{ symbols}$$

   6.75°

   53.3

   (b) Two radios use oscillators accurate to 2.5 ppm and transmit at 2.4 GHz. Find the largest frequency offset between them. Can a 4 kbaud QPSK Costas Loop lock to it? Can a 1 Mbaud one?

   Each can be off by $2.5 \times 10^{-6} \times 2.4$ GHz $= 6$ kHz, in opposite directions:

   $$\Delta f_{\text{max}} = 2 \times 6 \text{ kHz} = 12 \text{ kHz}$$

   At 4 kbaud the loop pulls in from about 375 Hz, 32 times too little. At 1 Mbaud the range scales to $375 \times 250 \approx 94$ kHz, which covers 12 kHz.

   12 kHz

   4 kbaud: no

   1 Mbaud: yes

   (c) In words, explain why an 8PSK Costas Loop pulls in from a smaller offset than a BPSK loop with the same bandwidth, and what the 4 kbaud receiver in part (b) needs before its Costas Loop.

   The phase detector cannot tell one constellation point from another, so it can only measure a phase error up to half the spacing, $180^\circ/M$: 90° for BPSK but 22.5° for 8PSK. A spinning constellation leaves the detector's range faster when M is larger, so the 8PSK loop has less time to correct and locks only from a smaller offset. The 4 kbaud receiver needs a coarse frequency correction first, such as an FLL Band-Edge block or a frequency measured from a known preamble, to bring the offset inside the Costas Loop's range.

5. **QPSK or 8PSK?** A link must carry 12 kb/s in a channel 6.25 kHz wide, with RRC pulses of roll-off $\alpha = 0.35$ and differential coding. Lesson 15 measured the E_b/N_0 the differential link needs for a bit error rate of 10^{-4}: 8.9 dB for QPSK and 12.4 dB for 8PSK.

   (a) Find the symbol rate and the occupied bandwidth with QPSK and with 8PSK. Which fits the channel?

   QPSK: $R_s = 12/2 = 6$ kbaud, $B = 1.35 \times 6 = 8.1$ kHz

   8PSK: $R_s = 12/3 = 4$ kbaud, $B = 1.35 \times 4 = 5.4$ kHz

   Only 8PSK fits in 6.25 kHz.

   QPSK: 8.1 kHz

   8PSK: 5.4 kHz

   (b) The bit rate is the same either way. How much more received power does the 8PSK link need for a bit error rate of 10^{-4}?

   At the same bit rate the received power is proportional to E_b, so the difference in E_b/N_0 is the difference in power:

   $$12.4 - 8.9 = 3.5 \text{ dB}$$

   3.5 dB

   (c) In words, state which modulation you would choose and what the link budget of Lesson 2 has to show before you commit to it.

   8PSK, because QPSK does not fit the channel at 12 kb/s. The link budget has to show at least 12.4 dB of E_b/N_0, or 17.2 dB of E_s/N_0, plus an allowance for the real radio and a fade margin, which is 3.5 dB more than QPSK would have needed.